3. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
knsat@SRITHIKA: ~

 GNU nano 8.7.1                                                                    states
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child Stage 1 - PID: %d, PPID: %d\n", getpid(), getppid());
        sleep(5);
        printf("Child Stage 2 - Waking up\n");
    } else if (pid > 0) {
        printf("Parent Stage 1 - PID: %d\n", getpid());
        wait(NULL);
        printf("Parent Stage 2 - Child done\n");
    }
    return 0;
}
```

```
knsat@SRITHIKA:~$ nano states.c
knsat@SRITHIKA:~$ gcc states.c -o states
knsat@SRITHIKA:~$ ./states
Parent Stage 1 - PID: 3532
Child Stage 1 - PID: 3533, PPID: 3532
Child Stage 2 - Waking up
Parent Stage 2 - Child done
knsat@SRITHIKA:~$ ps -l
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000     745     738  0  80   0 -  1567 do_wai pts/0    00:00:00 bash
0 R  1000    3543     745 99  80   0 -  1798 -      pts/0    00:00:00 ps
```